Module 0 Homework 4 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 79,992 of 99,990 points. **80%**

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1 Perfect square numbers have which possible last digits?

Your Answer	$1, 4, 9, 6, 5, 0$
Correct Answer	$1, 4, 9, 6, 5, 0$

Explanation

Solution. This question is very similar to the one we saw in the lesson! The last digit of a number is the only thing that determines the last digit of the square of the number. For example, if we wanted to know 687^2, this would be the same as finding $(68 \times 10 + 7)^2$, and we know that the term 68×10 multiplied by anything will end in a zero, so we can just ignore that term and look only at 7^2.

Since all that matters is the last digit, that means if we check the first ten positive perfect squares, we'll get all the possible last digits. Listing them out:

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

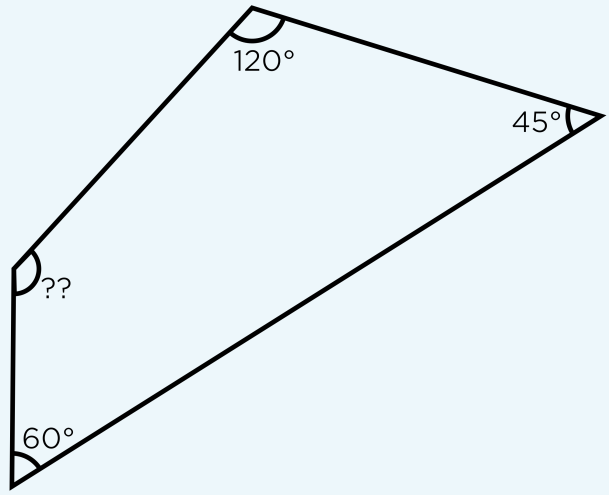
$$9^2 = 81$$

$$10^2 = 100$$

Then the possible last digits are $1, 4, 9, 6, 5, 6, 9, 4, 1,$ and 0.

Math Note. We saw back in the lesson that if a number ends in $2, 3, 7,$ or 8, it can't be a perfect square. This eliminates a lot of the answers for us!

2



How many degrees are in the marked angle?

Your Answer	[no answer given]
Correct Answer	135

Explanation

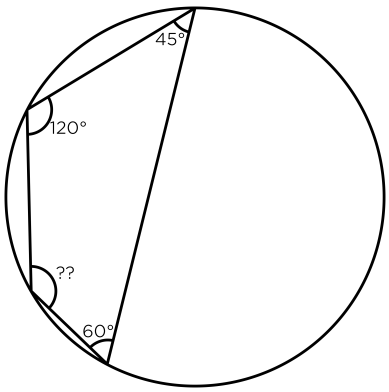
Solution. As we saw in the lesson, if we connect a pair of opposite *vertices* (corners), the quadrilateral can be cut into two triangles. Then the sum of the interior angles is $180^\circ + 180^\circ = 360^\circ$ degrees. The angle that we want is then

$$360^\circ - 60^\circ - 45^\circ - 120^\circ.$$

A shortcut for the subtraction is to recognize that $60^\circ + 120^\circ = 180^\circ$, and if we subtract that from the 360° first, we get

$$\begin{aligned} &360^\circ - 60^\circ - 45^\circ - 120^\circ \\ &= 180^\circ - 45^\circ \\ &= 135^\circ. \end{aligned}$$

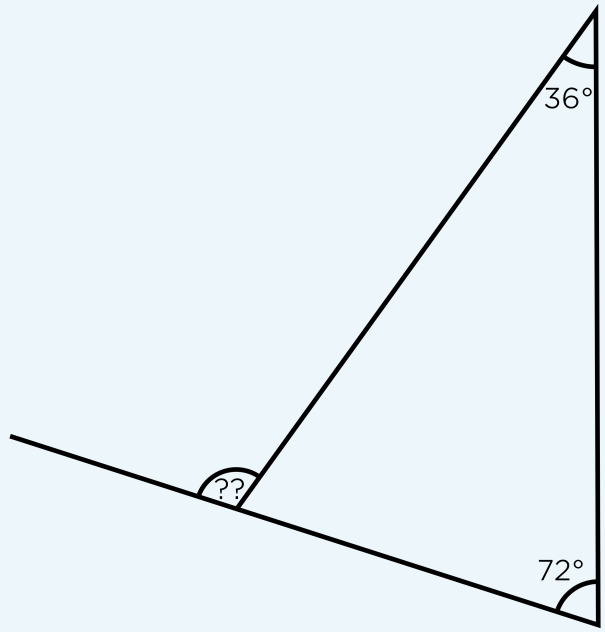
Math Note 1. This is actually a very special quadrilateral known as a cyclic quadrilateral, and it gets its name from the fact that all of its corners lie on a circle:



We'll learn in our Geometry course that this happens whenever the pairs of opposite angles of a quadrilateral add up to 180° degrees.

Math Note 2. The angles in this problem come up often, so it's good to get used to them!

3



How many degrees are in the marked angle?

Your Answer	[no answer given]
Correct Answer	108

Explanation

Solution 1. Remember from the lesson that the interior angles of a triangle add up to 180° degrees. We then know that the triangle has interior angles of 36°, 72°, and $180^\circ - 72^\circ - 36^\circ = 72^\circ$ degrees. This third interior angle, 72° makes a line with the angle we want, so the angle that we are looking for is

$$180^\circ - 72^\circ = 108^\circ.$$

Solution 2. You can sidestep some of the arithmetic above by remembering that an exterior angle of one corner of a triangle is equal to the sum of the two other angles of the triangle opposite to it. This is a neat fact, and we can trace it to the fact that angles on a line add up to 180° and the angles of a triangle also happen to sum to 180°. Well, then, the angle that we want is simply

$$36^\circ + 72^\circ = 180^\circ.$$

Math Note. This triangle is known as the golden triangle! We'll see it again later on.

4 What is $\left(100-29\right)+\left(382382-100\right)+\left(129-382382\right)$?

Your Answer	100
Correct Answer	100

Explanation

Solution 1. It's a bit nasty, but you can use brute force to compute the value of each set of parentheses and then add:

$$\left(100-29\right)+\left(382382-100\right)+\left(128-382382\right)=$$

$$71+382282-382254=$$

$$71+29=$$

$$=100$$

Solution 2. The niftier way is to regroup our sums to make this question much easier:

$$\left(100-29\right)+\left(382382-100\right)+\left(129-382382\right)=$$

$$\left(100-100\right)+\left(382382-382382\right)+\left(129-29\right)=$$

$$0+0+100=$$

$$=100$$

5 The sum of the interior angles in any polygon is always a multiple of which of these numbers?

✓	**Your Answer**	180
	Correct Answer	180

Explanation

Eyeball it. The numbers 101, 53, and 107 look suspicious, don't they? They don't seem like they will easily multiply with other numbers to get "nice" numbers ending in 0, and this is because 101, 53, and 107 are all prime, so they likely don't have prime factors in common with the number that is the sum of the interior angles of a polygon. Now, we can busy ourselves with looking at just 180° and 360°. The question is talking about polygons, which means any shape with three or more sides, and this includes triangles. Sometimes, one way of looking at a problem and testing a solution is to consider the extreme case. So if we consider a triangle with three sides, we know that its angles sum to 180°, which isn't a multiple of 360°. Our answer is 180°.

General polygons. In the lesson, we found that we can cut any polygon into triangles by choosing one corner to draw diagonal lines from, and drawing $n-3$ lines to the $n-3$ other corners of the polygon. (You could also draw triangles which all come out of the center of the polygon, but you then have to be careful to subtract the sum of all the central angles, which conveniently is 360°.)

Note: This works even if the polygon's sides are of different length, and even if the triangles formed are of different shapes, which is nice!

We know that the sum of the interior angles of the polygon will be some multiple, then, of a triangle's interior angle sum, in particular,

$$180^\circ \times \left(n-3\right).$$

Our answer, then, is 180°.

Note: It's interesting to notice that when n, the number of sides of the polygon, is odd, then the interior angle sum of the polygon is a multiple of 360°.

6 What is $\frac{5}{14} - \frac{1}{2} + \frac{1}{7}$?

Your Answer 0

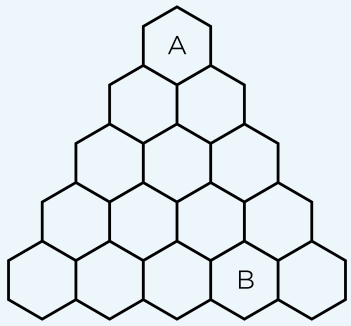
Correct Answer 0

Explanation

The idea here is to write $\frac{5}{14}$ as the difference of two fractions, and we suspect that our two fractions might have 2 and 7 as their denominators, since $2 \times 7 = 14$. This was the crux of the telescoping sum example in the lesson.

$$\begin{aligned} &\frac{5}{14} - \frac{1}{2} + \frac{1}{7} \\ &= \frac{1}{2} - \frac{1}{7} - \frac{1}{2} + \frac{1}{7} \\ &= 0. \end{aligned}$$

7

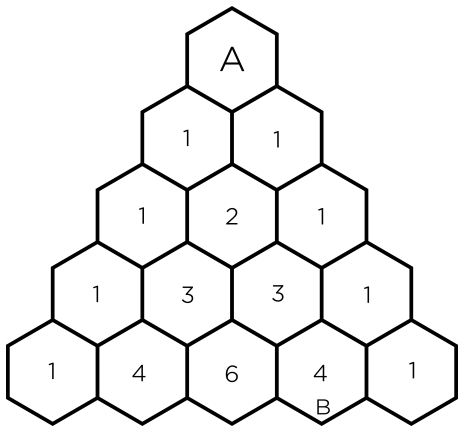


How many different paths are there from A to B if you can only go from one hexagon to a neighboring hexagon that is one of the two below it?

Your Answer	4
Correct Answer	4

Explanation

Solution 1. Since you can't move sideways, the only way to get to B is to have first come to one of the two hexagons that touch B's top corner. We can start from the top row and label each hexagon with the number of ways to reach it, like in the diagram below: must contain a factor of 7. The smallest integer containing a factor of 7 is 7. This satisfies the condition on *b*, that it has a remainer of 1 when divided by 3, so the answer is 7.



Actually, a neat simplification is to only fill out the right-most two hexagons in any row.

Solution 2. Another nice way to think about this problem is to "look at it a different way," in particular, by tilting your head to the right until the right edge looks horizontal. Then, to get to B from A, we just move 3 steps to the right and 1 step down, not necessarily in that order. We just need to make 4 steps, and there are 4 ways to choose which of those steps is the "down" step, so there are 4 ways in total to reach B.

Math note. The answer is 4 choose 1!

8 What are the last two digits of 201912^2?

✓	**Your Answer**	44
	Correct Answer	44

Explanation

The number 201912 can be written as

$$100 \times 2019 + 12,$$

so

$$201912^2 = \left(100 \times 2019 + 12\right)^2$$

$$= 2019^2 \times 100^2 + 12 \times 100 \times 2 + 12^2$$

$$= 100 \times \text{stuff} + 12^2$$

which has the same last two digits as $12^2 = 144$, so our answer is 44.

9 What is the last digit of 271828^6?

✓	**Your Answer**	4
	Correct Answer	4

Explanation

Solution. When calculating the final digit of a large power, we only need to look at the final digit of our original number! So, the last digit of 271828^6 is the same as the last digit of 8^6.

Instead of fully calculating out 8^6, we can just look at the final digit of each step of our calculation. Let's use the fact that $8^6 = (8^3)^2$. Then $8^3 = 8 \times 8^2 = 8 \times 64$, whose last digit is the same as the last digit of $8 \times 4 = 32$, which ends in 2. That was only 8^3, so we need to square our digit 2 again to get the last digit of 8^6, which will be $2^2 = 4$.

Math Note 1. Did you notice that the last digit of powers of 8 follow a cycle, and they go through $8, 4, 2$ and 6? This is actually true for the powers of any digit, and it can be used to save time on questions like this one!

Math Note 2. Does 271828 look familiar to you? You may remember from the lesson that 2.71828 is the first six digits of a very important mathematical constant, e.

Fun aside. Find out the fun connection between the mathematical constant, e, and expii!

10 What is $\frac{6}{17} \times \frac{7}{5} \times \frac{8}{3} \times \frac{5}{6} \times \frac{3}{7}$ expressed in simplest fractional form?
Your answer should be a fraction without spaces, e.g. 1/2.

✓	**Your Answer**	8/17
	Correct Answer	8/17

Explanation

Solution 1. Brute force it. Multiply the numerators on top to get

$$6 \times 7 \times 8 \times 5 \times 3 = 5040.$$

Multiply the denominators on the bottom to get

$$17 \times 5 \times 3 \times 6 \times 7 = 10710.$$

Simplifying, $\frac{5040}{10710}$ reduces to $\frac{8}{17}$.

Solution 2. In the same way that we've rearranged sums to make them easier to calculate, we can rearrange the numerators and denominators of the fractions in this product. Rewriting gives us

$$\frac{8}{3} \times \frac{3}{7} \times \frac{7}{5} \times \frac{5}{6} \times \frac{6}{17}.$$

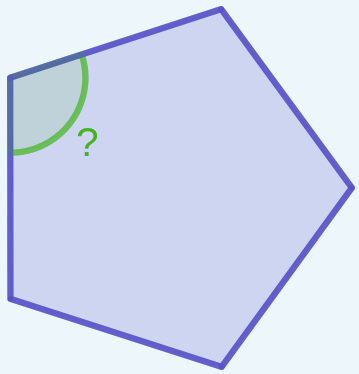
Now it is easier to see that each numerator cancels the denominator before it, simplifying to $\frac{8}{17}$.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

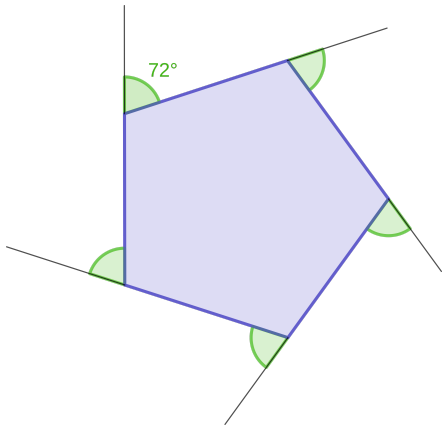
11



How many degrees are there in the interior angle of a regular pentagon?

Your Answer	[no answer given]
Correct Answer	108

Explanation



Let's use the interesting fact that the exterior angles of the pentagon (angles that you get by extending each side out past its corner) all add up to 360°. (Remember the example of the ant crawling along the sides, turning at each corner, with his turns adding up to a full revolution?) Since there are five of these exterior angles, we can find the measure of each one: $\frac{360^\circ}{5} = 72^\circ$. We want the measure of just one interior angle, which is $180^\circ - 72^\circ = 108^\circ$.

Math note. The fact we used isn't just true for a pentagon; it's true for any polygon, even one with 100 sides! (Then, each exterior angle would have measure 3.6°, and it would look pretty much like a circle.)

12 What is the remainder when 13^3 is divided by 8?

Your Answer	[no answer given]
Correct Answer	5

Explanation

It's helpful to imagine that there are multiples of 8 inside 13^3 which we can remove at each step of our calculation, so we can just think about the remainders that are left behind. Let's think of 13 as $8+5$, so

$$\left(8+5\right)^2 = 8 \times \text{stuff} + 5^2,$$

where a is an integer whose value doesn't matter. The remainder after dividing by 8 will be the same as the remainder after dividing 5^2 by 8, which is 25 divided by 8, which is 1.

We just found the remainder of 13^2, but we really want the remainder of 13^3, so we can write

$$13^3 = 13^2 \times 13$$

which has the same remainder as

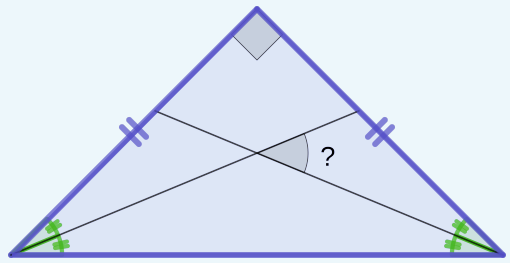
$$1 \times 13$$

which has the same remainder as

$$1 \times \left(8+5\right)$$
$$= 5.$$

So, 5 is our answer.

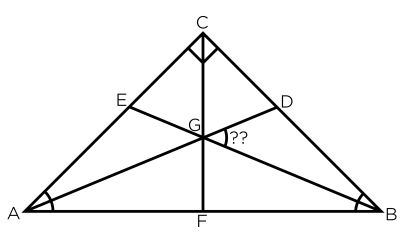
13



This is a special right triangle with two legs that are the same length. The two lines crossing through the triangle are *bisectors* (they split the angles they go through into half). How many degrees are in the marked angle?

✕	**Your Answer**	[no answer given]
	Correct Answer	45

Explanation



Solution 1. When a questions asks for an angle, a great idea is to start hunting for a triangle that contains that angle, and it's even better if we already know the measures of its other angles. Let's look at $\triangle BGD$ in the diagram above. We actually already know $\angle DBG$, because $\overline{BE}$ is an angle bisector of $\angle ABD$ (which we know is 45°, since the large triangle has equal legs), so that means that $\angle DBG = \frac{45^\circ}{2}$. One trick is to keep our angles in terms of 45° rather than calculating them out as decimals, because we will find that some of our numbers will cancel out nicely!

We don't yet know $\angle BDG$, though, so we need to start being creative. Let's look for more triangles that can give us $\angle BDG$. There are two triangles that we can use, $\triangle ABD$ and $\triangle ACD$.

In triangle $\triangle ABD$, we know $\angle DAB$ is again $\frac{45^\circ}{2}$ because $\overline{AD}$ is an angle bisector. We also know that $\angle ABD = 45^\circ$, so

$$\begin{aligned}\angle BDG &= 180^\circ - 45^\circ - \frac{45^\circ}{2} \\ &= 90^\circ + \frac{45^\circ}{2}.\end{aligned}$$

If we use triangle $\triangle ACD$, then we know $\angle CAD = \frac{45^\circ}{2}$ and $\angle ACD = 90^\circ$, so we can use that to find $\angle ADC$.

$$\begin{aligned}\angle ADC &= 180^\circ - 90^\circ - \frac{45^\circ}{2} \\ &= 90^\circ - \frac{45^\circ}{2}\end{aligned}$$

Since $\angle ADC$ and $\angle BDG$ are *supplementary* (add up to a straight line),

$$\angle BDG = 90^\circ + \frac{45^\circ}{2}.$$

Now that we know $\angle BDG$,

$$\begin{aligned}\angle BGD &= 180^\circ - \angle DBG - \angle BDG \\ &= 180^\circ - \frac{45^\circ}{2} - \left(90^\circ - \frac{45^\circ}{2}\right) \\ &= 45^\circ.\end{aligned}$$

Solution 2. We can also look at triangle $\triangle ABG$, because one of its angles, $\angle BGA$, is supplementary to the angle we want, $\angle BGD$. It has two angles measuring $\frac{45^\circ}{2}$, so

$$\begin{aligned}\angle BGA &= 180^\circ - \frac{45^\circ}{2} - \frac{45^\circ}{2} \\ &= 180^\circ - 45^\circ \\ &= 135^\circ\end{aligned}$$

Then

$$\begin{aligned}\angle BGD &= 180^\circ - \angle BGA \\ &= 180^\circ - 135^\circ \\ &= 45^\circ.\end{aligned}$$

14 Positive integer a has remainder 1 when divided by 7, while positive integer b has remainder 1 when divided by 3.
What is the smallest possible value of b that makes $a \times b$ always a multiple of 7?

Your Answer	[no answer given]
Correct Answer	7

Explanation

Solution 1. Since a is a positive integer that has remainder 1 when divided by 7, it is one more than a multiple of 7, and looks like

$$7 \times c + 1,$$

where the exact value of the integer c doesn't matter. In the same way, b is one more than a multiple of 3, and looks like

$$3 \times d + 1,$$

where we don't yet know the value of d. If we multiply a and b together, we get:

$$a \times b =$$

$$\left(7c + 1\right) \times \left(3d + 1\right) =$$

$$21 \times cd + 7c + 3d + 1.$$

The first two terms are multiples of 7, so the whole thing will be a multiple of 7 if $3d + 1$ is also a multiple of 7. The smallest positive multiple of 7 is 7, so let's check to see if d is an integer when $3d + 1 = 7$. To do this, just solve for d :

$$3d + 1 = 7$$

$$3d = 7 - 1$$

$$3d = 6$$

$$d = 2.$$

Remember, we're looking for the value of b, not d, so

$$b = 3d + 1,$$

$$b = 7.$$

It's interesting to see that none of this depends on the value of c or a.

Solution 2. It's possible to do this problem without very much math! Integer a has a remainder of 1 when divided by 7, so we know that it contains no factors of 7. For $a \times b$ to be a multiple of 7, b must contain a factor of 7. The smallest integer containing a factor of 7 is 7. We needed b to have a remainder of 1 when divided by 3, and this works for $b = 7$, so the answer is 7.

15 Which of the following is the same as the fraction shown below?

$$\frac{6\sqrt{6}-5\sqrt{5}}{30\sqrt{30}}$$

Your Answer [no answer given]

Correct Answer $\frac{1}{5\sqrt{5}}-\frac{1}{6\sqrt{6}}$

Explanation

We can write $\sqrt{30}$ as $\sqrt{5}\times\sqrt{6}$, which lets us cancel factors from the top and bottom:

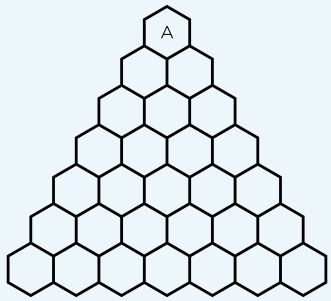
$$\frac{6\sqrt{6}-5\sqrt{5}}{30\sqrt{6}\sqrt{5}}$$

$$=\frac{6\sqrt{6}}{30\sqrt{6}\sqrt{5}}-\frac{5\sqrt{5}}{30\sqrt{6}\sqrt{5}}$$

$$=\frac{6}{30\sqrt{5}}-\frac{5}{30\sqrt{6}}$$

$$=\frac{1}{5\sqrt{5}}-\frac{1}{6\sqrt{6}}$$

16

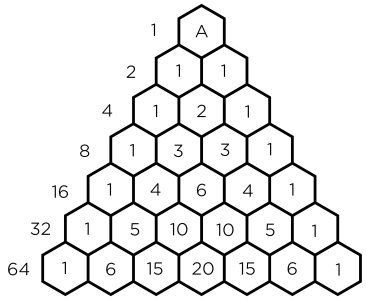


How many different paths are there from A to *any spot* on the bottom row if a bee can only go from one hexagon to a neighboring hexagon that is one of the two below it?

✕	**Your Answer**	[no answer given]
	Correct Answer	64

Explanation

Solution 1. Label each hexagon with the number of ways to get to it from A. The number of ways to reach a given hexagon is the number of ways to get to the hexagon that is above-and-right plus the number of ways to get to the hexagon that is above-and-left. Working from the top to the bottom, we can fill out the numbers as shown and add the last row of numbers to get the answer, 64.



Solution 2. Let's look at the total number of paths from A to any row; let's call it the nth row. For every path to the $(n-1)^{\text{th}}$ row right above it, there are two ways to continue on to the nth row. In this way, every additional row has double the total number of paths to it compared to the previous row. The bottom row is 6 rows down, and we start with 1 path in the first row, so the total number of paths is $2^6 = 64$.

Math note. In general, the nth row of Pascal's triangle sums to $2^n - 1$!

17 What is $\frac{5}{14}+\frac{4}{77}-\frac{5}{66}+\frac{7}{78}+\frac{1}{182}$?

Your answer should be a fraction in simplest form, without spaces (e.g. 1/2).

✕	**Your Answer**	[no answer given]
	Correct Answer	3/7 | 3 / 7

Explanation

This question is a lot like the telescoping sum question that we saw in the videos. Let's rewrite each fraction as the difference of two other fractions and hope the sum telescopes!

$$\frac{5}{14}=\frac{1}{2}-\frac{1}{7}$$

$$\frac{4}{77}=\frac{1}{7}-\frac{1}{11}$$

$$\frac{5}{66}=\frac{1}{6}-\frac{1}{11}$$

$$\frac{7}{78}=\frac{1}{6}-\frac{1}{13}$$

and

$$\frac{1}{182}=\frac{1}{13}-\frac{1}{14}.$$

Then, our sum becomes

$$\begin{aligned}\frac{1}{2}-\frac{1}{7}+\frac{1}{7}-\frac{1}{11}-\frac{1}{6}+\frac{1}{11}+\frac{1}{6}-\frac{1}{13}+\frac{1}{13}-\frac{1}{14}\\ =\frac{1}{2}-\frac{1}{14}\\ =\frac{7}{14}-\frac{1}{14}\\ =\frac{6}{14}\\ =\frac{3}{7}.\end{aligned}$$

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Which of the following is equivalent to

$$\frac{3\sqrt{3}-2\sqrt{2}}{6\sqrt{6}}+\frac{8-3\sqrt{3}}{24\sqrt{3}}+\frac{5\sqrt{5}-8}{40\sqrt{5}}+\frac{6\sqrt{6}-5\sqrt{5}}{30\sqrt{30}}?$$

Your Answer [no answer given]

Correct Answer $\frac{3\sqrt{3}-1}{6\sqrt{6}}$

Explanation

Let's try to look for a pattern in the terms of the expression. Since each numerator is written as a difference of two numbers, let's break each fraction up:

$$\frac{3\sqrt{3}-2\sqrt{2}}{6\sqrt{6}}+\frac{8-3\sqrt{3}}{24\sqrt{3}}+\frac{5\sqrt{5}-8}{40\sqrt{5}}+\frac{6\sqrt{6}-5\sqrt{5}}{30\sqrt{30}}$$

$$=\frac{3\sqrt{3}}{6\sqrt{6}}-\frac{2\sqrt{2}}{6\sqrt{6}}+\frac{8}{24\sqrt{3}}-\frac{3\sqrt{3}}{24\sqrt{3}}+\frac{5\sqrt{5}}{40\sqrt{5}}-\frac{8}{40\sqrt{5}}+\frac{6\sqrt{6}}{30\sqrt{30}}-\frac{5\sqrt{5}}{30\sqrt{30}}$$

This is easier than just adding all of the original terms together directly, because there are many common factors.

Now, let's cancel out factors:

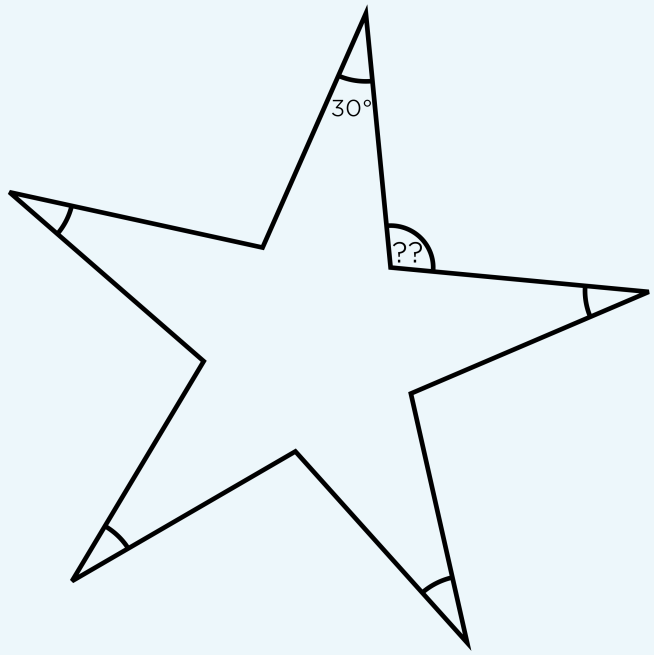
$$\frac{1}{2\sqrt{2}}-\frac{1}{3\sqrt{3}}+\frac{1}{3\sqrt{3}}-\frac{1}{8}+\frac{1}{8}-\frac{1}{5\sqrt{5}}+\frac{1}{5\sqrt{5}}-\frac{1}{6\sqrt{6}}$$

Now all but the first and last terms disappear because they telescope, giving:

$$\frac{1}{2\sqrt{2}}-\frac{1}{6\sqrt{6}}$$

$$= \frac{3\sqrt{3} - 1}{6\sqrt{6}}$$

19



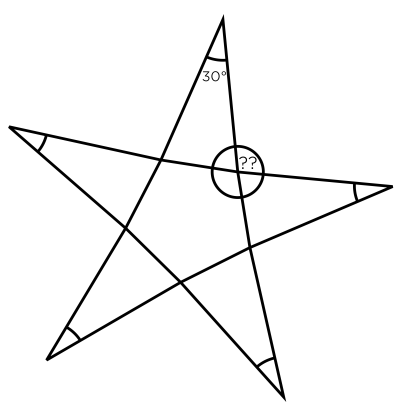
How big is the marked angle, in degrees?

Your Answer	[no answer given]
Correct Answer	102

Explanation

This star has a regular pentagon at its center and five triangles attached to each edge of the pentagon. Let's look at all of the angles that, together with the angle that we want, surround one of the corners of the pentagon:

$$2 \times \left(\text{triangle } \angle\right) + \left(\text{interior } \angle \text{ of pentagon}\right) = 360^\circ.$$



We can get the internal angle of the triangle using the given 30° angle.

triangle $\angle = \frac{180^\circ - 30^\circ}{2} = 75^\circ$.

From an earlier problem, we know that the interior angle of a pentagon is 108°. Then the angle we want is

$$360^\circ - \left(2 \times 75\right) - 108^\circ$$

$$= 102^\circ.$$

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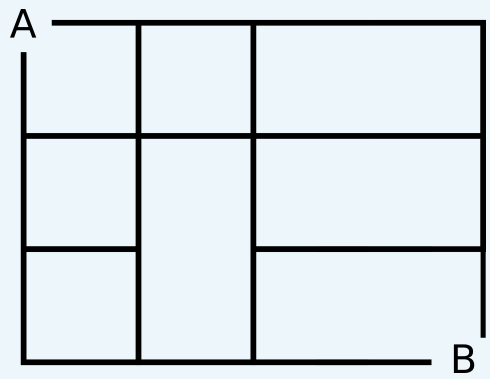
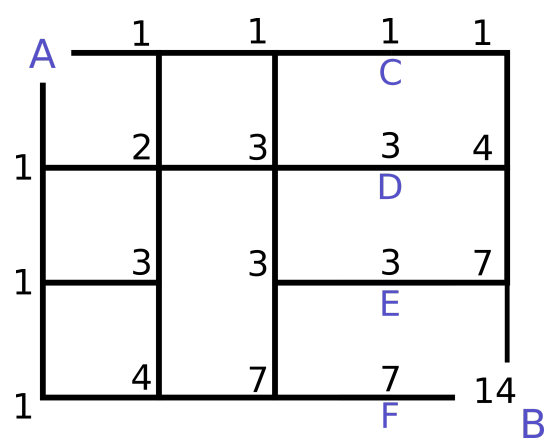


How many different paths are there from A to B if you can only go down and to the right?

Your Answer	[no answer given]
Correct Answer	14

Explanation



Solution 1. In order to get to any intersection point, we must have come from either above or from left. Then the number of ways to get to any intersection is simply the number of paths to the intersection directly above plus the number of paths to the intersection directly to the left. Let's fill out the grid as shown from the top left to the bottom right. Start from A, and fill in the intersections points that are nearest to A, working your way towards B. As you can see, the number of ways to get to B is 14.

Solution 2. Since there are no vertical segments down the middle, each path must pass through exactly one of C, D, E, or F. Let's look at these four cases separately and then add up all of their ways at the end.

ways from A to B = $\left(\text{\# paths from } A \text{ to } C\right) \times \left(\text{\# paths from \# } C \text{ to } B\right)$

This also works for paths going through D, E, and F, also.

Actually, there is only one way to get from any of C, D, E, and F to B, so the total number of ways to get to B is just the sum of the ways to get to C, D, E, and F, which equals $1 + 3 + 3 + 7 = 14$.

Summary

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